

Hybrid CNN – LSTM – SVM model for advanced brain tumor classification

Final Review — DL Tumor Classification
Five backbones · leakage audit · frozen external validation · web stress test · explainability

Guide
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Final Project Overview

What we built — and what we found

Built

- Hybrid CNN → LSTM → Dense128 → Scaler → PCA64 → linear SVM
- 5 independent backbones (VGG16/19, ResNet50/152, EfficientNet-B3)
- Phase 1, 2A (5 runs), 2B (35 runs + 70 downstream evals)
- Leakage audit of 7,023 images
- 15 pipelines frozen + SHA-256 hashed
- BDNeuro external (2,644) + web stress test (40)
- Grad-CAM, final-SVM occlusion, Streamlit prototype

Final representative: ResNet152 Phase 2A (epoch 10)

Evaluation	Accuracy
Internal validation	98.60%
Internal test*	98.70%
material_clean test*	98.869%
BDNeuro full (2,644)	88.31%
BDNeuro balanced (1,648)	90.47%
Web-40 stress test	**37.5%**

** original split, later shown to contain duplicate leakage*

High internal accuracy ≠ robustness to domain shift

Introduction & Motivation

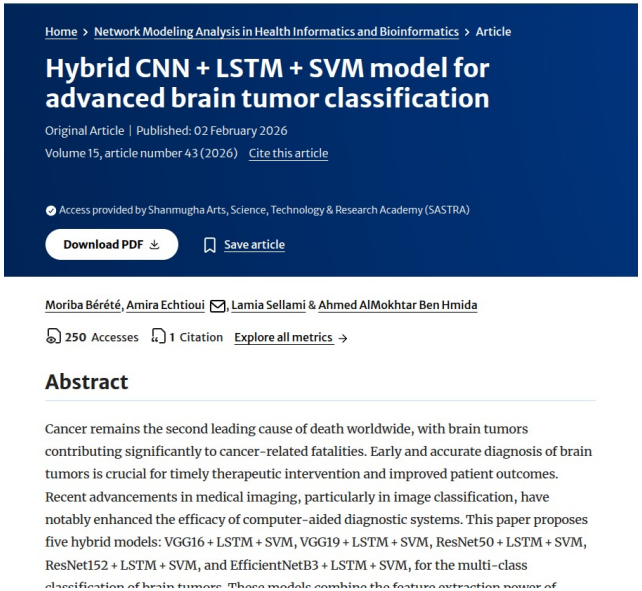
- Glioma, meningioma and pituitary tumours are frequent brain tumours; accurate identification supports treatment planning.
- Manual MRI reading is time-consuming, expert-dependent and subject to inter-observer variability.
- Deep learning reports ~98–99% accuracy on public brain-MRI datasets.
- But high accuracy alone does not establish reliability — **data leakage, dataset shift and model opacity** must be examined.
- Our final project examined all three — and all three changed the interpretation.

Glioma | Meningioma | No Tumour | Pituitary

Base Paper

Moriba Bérété, Amira Echtioui, Lamia Sellami & Ahmed AlMokhtar Ben Hmida, “Hybrid CNN+LSTM+SVM model for advanced brain tumor classification”, Network Modeling Analysis in Health Informatics and Bioinformatics, Springer, Volume 15, Article 43, 2026.

DOI: 10.1007/s13721-025-00718-8
Indexed in: Scopus / SCI / SCI-E
Year: 2026



Home › Network Modeling Analysis in Health Informatics and Bioinformatics › Article

Hybrid CNN + LSTM + SVM model for advanced brain tumor classification

Original Article | Published: 02 February 2026
Volume 15, article number 43 (2026) Cite this article

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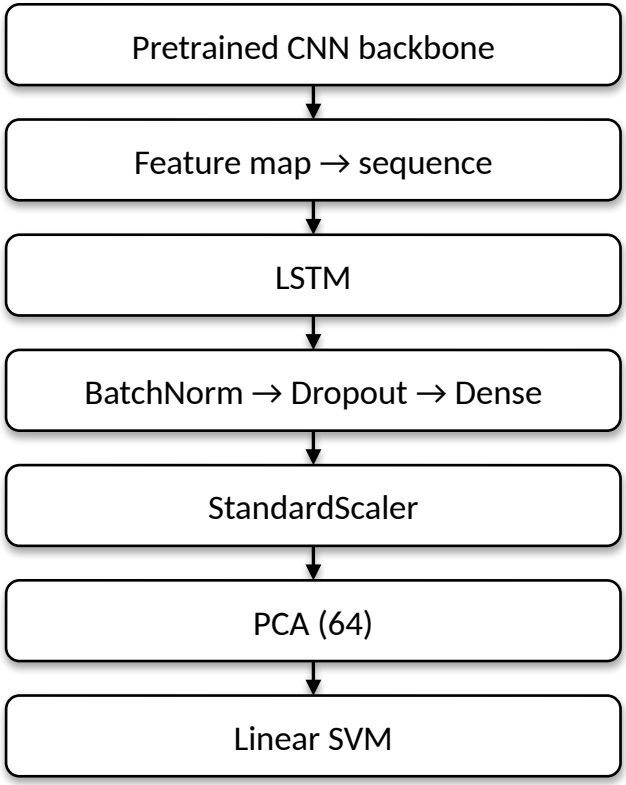
Moriba Bérété, Amira Echtioui, Lamia Sellami & Ahmed AlMokhtar Ben Hmida

250 Accesses 1 Citation Explore all metrics →

Abstract

Cancer remains the second leading cause of death worldwide, with brain tumors contributing significantly to cancer-related fatalities. Early and accurate diagnosis of brain tumors is crucial for timely therapeutic intervention and improved patient outcomes. Recent advancements in medical imaging, particularly in image classification, have notably enhanced the efficacy of computer-aided diagnostic systems. This paper proposes five hybrid models: VGG16 + LSTM + SVM, VGG19 + LSTM + SVM, ResNet50 + LSTM + SVM, ResNet152 + LSTM + SVM, and EfficientNetB3 + LSTM + SVM, for the multi-class

Base Paper – Method & Reported Findings



RMSprop results as reported [1, Table 3]

Model (RMSprop)	Accuracy
VGG16	94%
VGG19	96%
ResNet50	99%
ResNet152	**99%**
EfficientNetB3	98%

Protocol (as reported)
5 backbones × 4 optimizers · 30 epochs · batch 32 · LR 1e-4 · dropout 0.3
Not reported: duplicate check · external dataset · implemented XAI

Identified Limitations → Our Solution

Limitation identified in literature	Our solution	Phase
High accuracy without duplicate / leakage checks [1]	Cross-split hash + perceptual audit; 3 leakage-aware test policies	2D ✓
Single-dataset evaluation [1, 9, 12]	15 frozen pipelines on an overlap-audited external cohort, no tuning	2C ✓
Early stopping hides convergence	Fixed 30 epochs; best-within-30 and epoch-30 states	2A ✓
Search-based hyperparameter choice [14]	One-factor study: LR, LSTM size, dropout	2B ✓
Ensembles / single backbones [11, 13, 15]	Five independent backbones, one protocol	1, 3 ✓
Grad-CAM cannot explain an SVM [10]	Grad-CAM (neural head) + final-SVM occlusion	4 ✓
Heatmaps over-read as localization [12]	Rated as coarse contribution maps — no segmentation claim	4 ✓
(added) Severe domain shift untested	Source-verified web-40 stress test, locked before inference	2C+ ✓

Problem Statement & Objectives

To build and rigorously evaluate a hybrid CNN–LSTM–PCA–SVM framework for four-class brain MRI classification, examining backbone choice, training behaviour, data integrity and generalization — as a research framework, not a clinical diagnostic device.

Objective	Where	Status
Reconstruct & adapt the base-paper pipeline (deviations documented)	Ph 1	✓ Complete
Compare five backbones under one protocol	Ph 1, 3	✓ Complete
Training behaviour & hyperparameter sensitivity	Ph 2A, 2B	✓ Complete
Verify data integrity; leakage-aware evaluation	Ph 2D	✓ Complete
Overlap-audited external validation, models frozen	Ph 2C	✓ Complete
Explainability: Grad-CAM + occlusion	Ph 4	✓ Complete
Research & educational prototype	Ph 5	✓ Complete

Original Roadmap vs Final Status

PHASE 1 — BASELINE SYSTEM ✓

- Brain MRI Dataset
- Dataset Preparation | Preprocessing
- CNN Feature Extractor
- Spatial Feature Sequence
- LSTM
- 128-D Deep Features
- StandardScaler
- PCA (64)
- Linear SVM
- Initial Evaluation + Neural Head

PHASE 2 — OPTIMISATION & VALIDATION ✓

- 2A 30-Epoch Protocol ✓
- 2B Hyperparameter Analysis ✓
- 2C External Validation ✓
- 2D Data Integrity / Robustness ✓ — leakage audit + 3 test policies
- + Web-40 stress test (beyond plan) ✓

PHASE 3 — COMPARATIVE ANALYSIS & FINAL MODEL SELECTION ✓

PHASE 4 — EXPLAINABILITY (Grad-CAM + occlusion) ✓

PHASE 5 — PROTOTYPE APPLICATION ✓

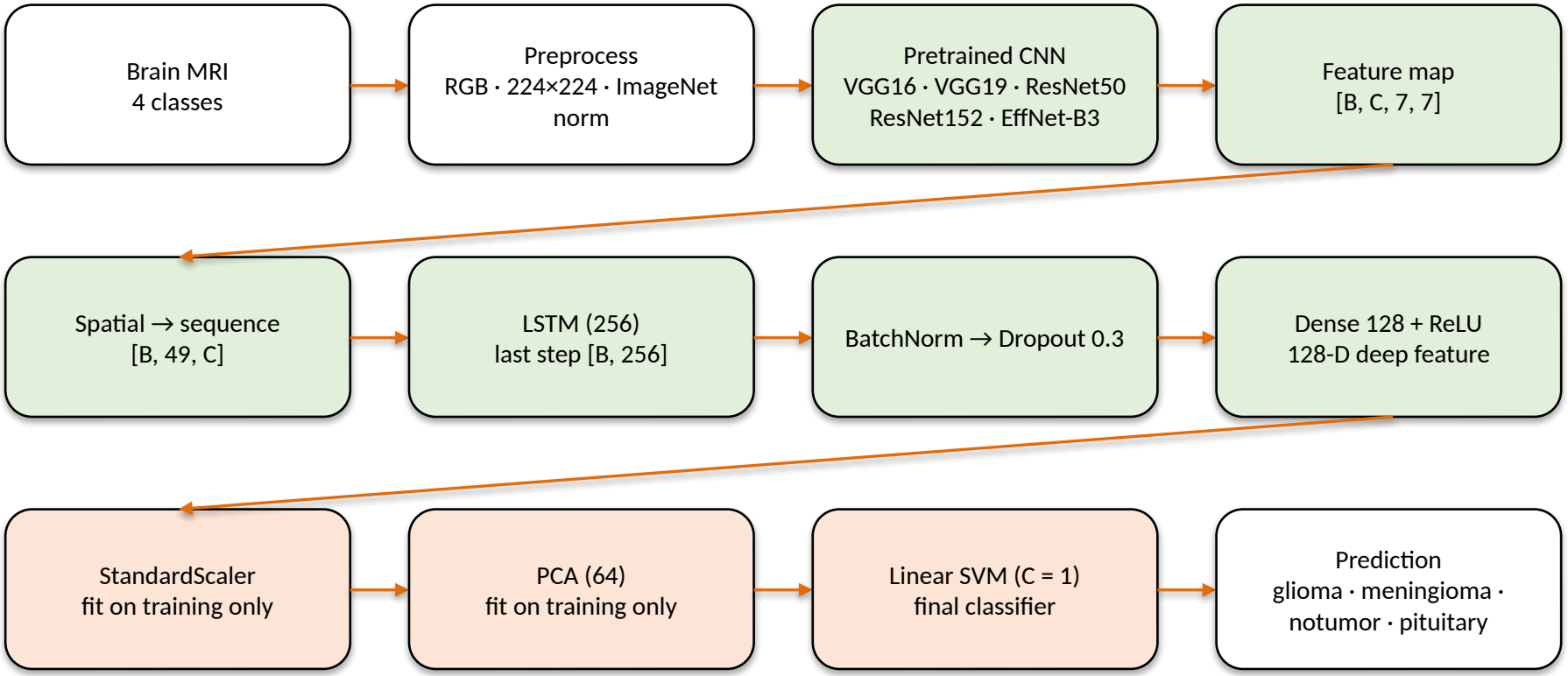
PHASE 6 — FINAL VALIDATION, REPORT & DEMONSTRATION ✓

✓ Completed

Added beyond plan

Final Project Architecture

One independent pipeline per backbone — not an ensemble



Neural head (training & comparison): Dense128 → Linear 4 logits · CrossEntropyLoss on raw logits · softmax only at inference. VGG16/19 CNN frozen; ResNet50/152 & EfficientNet-B3 fine-tuned.

Dataset, Preprocessing & Split

Class	Training	Testing	Total
Glioma	1,321	300	1,621
Meningioma	1,339	306	1,645
No tumour	1,595	405	2,000
Pituitary	1,457	300	1,757
Total	5,712	1,311	7,023

Split (seed 42, stratified)
Train 4,569 — fits CNN-LSTM, scaler, PCA, SVM
Validation 1,143 — checkpoint, state & config selection
Test 1,311 — final evaluation only; never used for selection

Every image
RGB conversion · resize 224 × 224 · ImageNet mean/std normalization

Training only (src/data_pipeline_torch.py L65–78)
H-flip p=0.5 · V-flip p=0.5 · ColorJitter(brightness 0.1, contrast 0.2, saturation 0.2, hue 0.1)
Validation / test / features / GUI: deterministic transform

Implementation Journey

TensorFlow → PyTorch, then a verified, step-wise build

28 Aug	TensorFlow on RTX 5070 → INVALID_PTX (sm_120); nightly abandoned
→	PyTorch 2.11.0 + cu128 under WSL2; CUDA verified
Late Aug	Seed-42 split saved; RGB · 224 · ImageNet norm; train-only augmentation
Early Sep	Step-wise tests: CNN map → sequence → LSTM → post-LSTM → full model → real batch → training step
Early Sep	First ResNet152 hybrid end-to-end (legacy, superseded)
6–12 Sep	Generic CNNLSTM for 5 backbones; VGG16 collapse (~28% val) → freeze VGG backbones

Phase 1 — Baseline Methodology

Backbone	CNN policy	Epochs run / max	Best epoch	PCA variance kept
VGG16	frozen	19 / 30	16	99.18%
VGG19	frozen	21 / 30	18	99.33%
ResNet50	fine-tuned	10 / 10	7	99.45%
ResNet152	fine-tuned	10 / 10	8	99.39%
EfficientNet-B3	fine-tuned	9 / 10	6	99.33%

- RMSprop · LR 1e-4 · batch 8 (drop_last) · LSTM 256 · Dense 128 · dropout 0.3 · seed 42 · CE on raw logits
- Best-validation checkpoint; early stopping patience 3 — unequal budgets (30 vs 10) motivated Phase 2A
- Deterministic 128-D features → StandardScaler → PCA 64 → linear SVM (fit on train only)

Phase 1 — Neural-Head Results

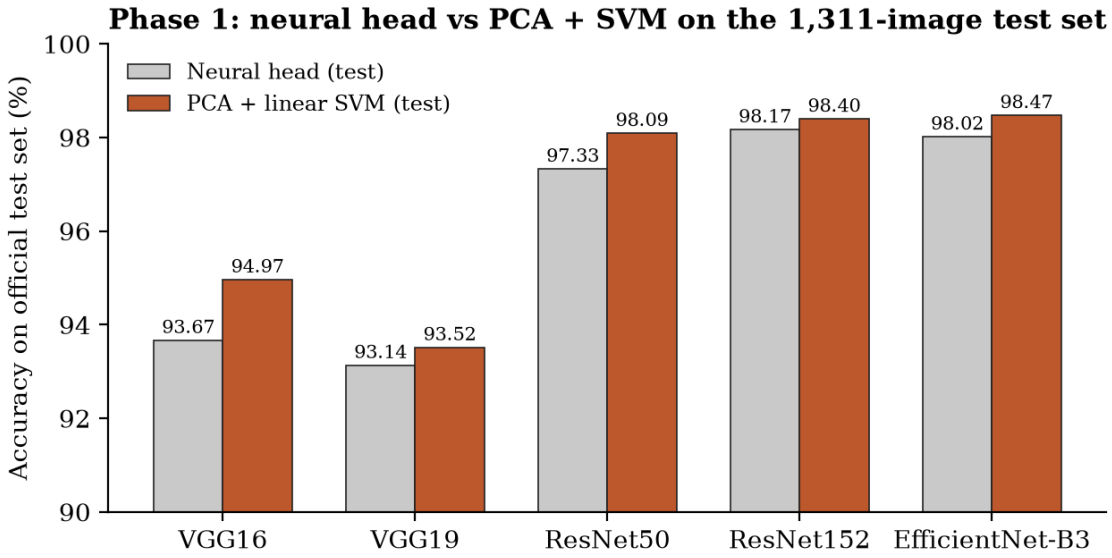
Before PCA/SVM: the CNN-LSTM's own 4-class head

Backbone	Best epoch	Validation acc	Test acc	Test macro F1
VGG16	16	94.58%	93.67%	93.32%
VGG19	18	94.58%	93.14%	92.72%
ResNet50	7	97.73%	97.33%	97.19%
ResNet152	8	98.25%	98.17%	98.06%
EfficientNet-B3	6	98.43%	98.02%	97.94%

Original 1,311-image test set — later shown to contain duplicate leakage (slide 17).

PCA + SVM vs Neural Head

Downstream-classifier comparison (not a PCA-only ablation)



Backbone	Δ test	Δ val
VGG16	+1.30 pp	+0.35 pp
VGG19	+0.38 pp	-0.09 pp
ResNet50	+0.76 pp	-0.17 pp
ResNet152	+0.23 pp	+0.00 pp
EfficientNet-B3	+0.46 pp	+0.09 pp

Test ↑ for all 5; validation mixed → no universal PCA claim

Duplicate / Leakage Audit (Phase 2D)

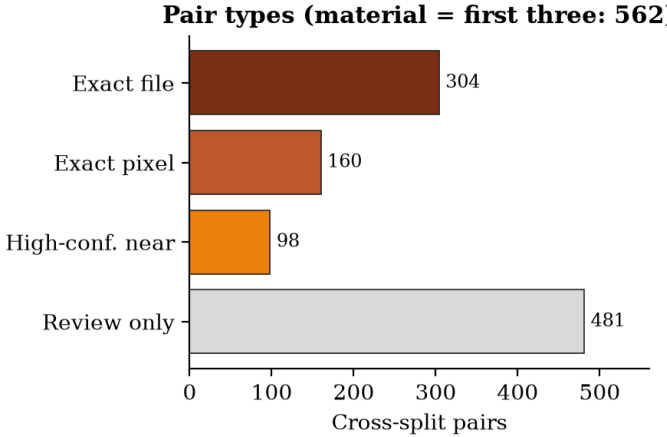
7,023 images · file SHA-256 · decoded-pixel SHA-256 · pHash/dHash + MAE/correlation

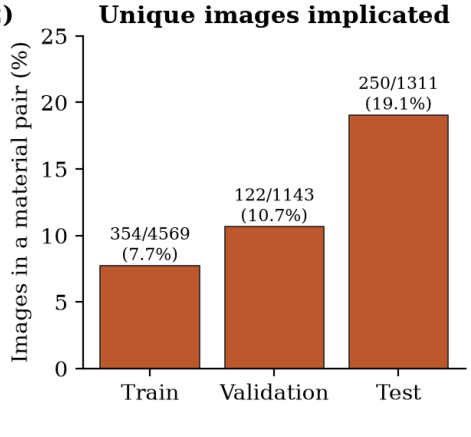
464
exact file / pixel pairs

98
high-confidence near pairs

562
material pairs

250
test images implicated (19%)





Status: MATERIAL_DUPLICATION
Unique images implicated: train 354 · val 122 · test 250
Test by class: gli 0 · men 109 · no 136 · pit 5
Patient-level independence: cannot be established

Leakage-Aware Internal Evaluation

Post-hoc re-scoring of saved models — no retraining, no refitting

Backbone	Original n=1,311	exact_clean n=1,095	material_clean n=1,061	material F1	conservative n=926
VGG16	94.97%	94.98%	94.816%	94.302%	94.06%
VGG19	93.52%	93.24%	93.025%	92.415%	92.01%
ResNet50	98.09%	98.36%	98.303%	98.134%	98.06%
ResNet152	98.40%	98.45%	98.398%	98.315%	98.16%
EfficientNet-B3	98.47%	98.63%	98.680%	98.511%	98.49%

Supported
Strong Phase 1 scores are not driven solely by duplicated test images ($|\Delta| < 0.5$ pp)

Not supported
material_clean does not repair training contamination · no patient-level independence

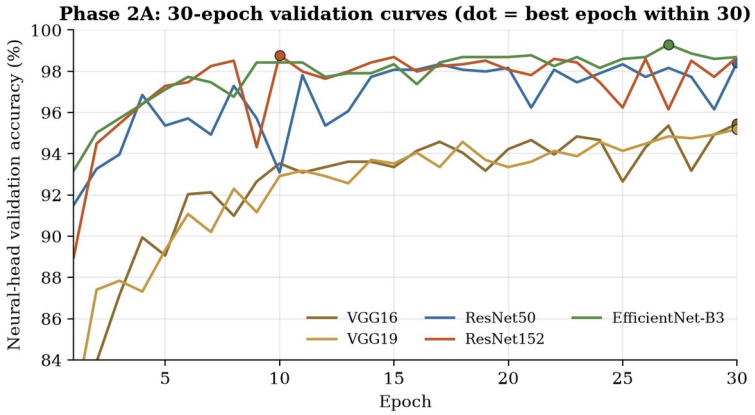
Phase 2A — 30-Epoch Standardized Protocol

- Why: early stopping hides convergence and late-epoch drift; Phase 1 budgets were unequal
- All five baselines: RMSprop · LR 1e-4 · LSTM 256 · dropout 0.3 · batch 8 · seed 42
- Exactly 30 / 30 epochs, no early termination
- Saved: best_val_within_30.pth and epoch_30_model.pth
- Both states → Scaler → PCA 64 → SVM: 10 evaluations
- Representative chosen by full-hybrid validation (ties → macro-F1 → best_val_within_30)

Technical note
VGG16, VGG19, ResNet50: best epoch = 30, so both states tie (7 distinct results of 10).
Frozen representative = best_val_within_30 at epoch 30 — not epoch_30_model.pth (files differ by hash).

Phase 2A — Results

Backbone	Representative state	SVM val	Test*	mat.-clean*
VGG16	best_val_within_30 · ep 30	95.888%	96.186%	95.853%
VGG19	best_val_within_30 · ep 30	95.626%	95.271%	95.099%
ResNet50	best_val_within_30 · ep 30	98.688%	98.932%	98.963%
ResNet152	**best_val_within_30 · ep 10**	**98.600%**	**98.703%**	**98.869%**
EfficientNet-B3	best_val_within_30 · ep 27	99.300%	99.314%	99.246%



- VGG16/19 improved until epoch 30 → Phase 1 early stopping halted them early
- ResNet152 peaked at epoch 10 (98.78% neural val) and stayed close (98.69% at 30)
- * contaminated original split; not used for any selection

Phase 2B — Hyperparameter Sensitivity Design

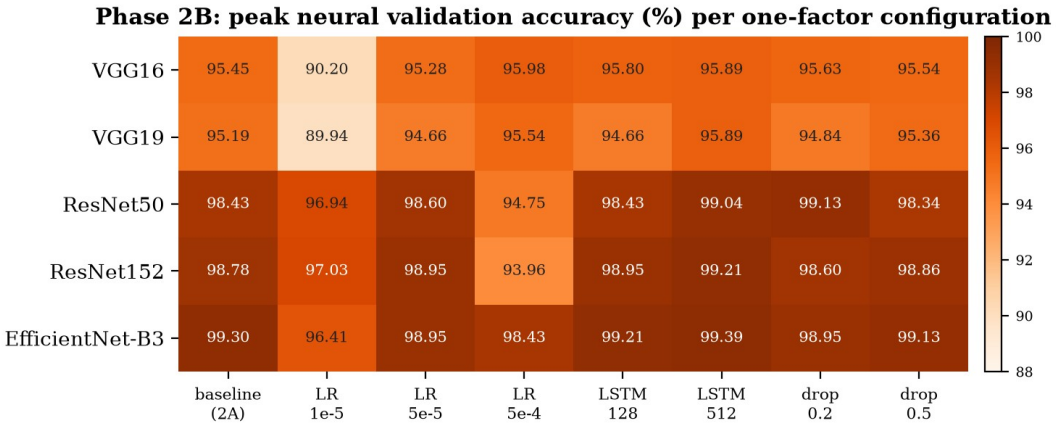
Run	LR	LSTM	Dropout
lr_1e-5	1e-5	256	0.3
lr_5e-5	5e-5	256	0.3
lr_5e-4	5e-4	256	0.3
lstm_128	1e-4	128	0.3
lstm_512	1e-4	512	0.3
dropout_0.2	1e-4	256	0.2
dropout_0.5	1e-4	256	0.5

5 × 7 = 35 runs
all 30 / 30 epochs
35 × 2 states = 70 downstream evals

Test set deliberately NOT evaluated
70 states compared → scoring test would make it a tuning set; selection is validation-only

One-factor-at-a-time around the Phase 2A baseline (LR 1e-4 · LSTM 256 · dropout 0.3); every other setting fixed.

Phase 2B — Findings



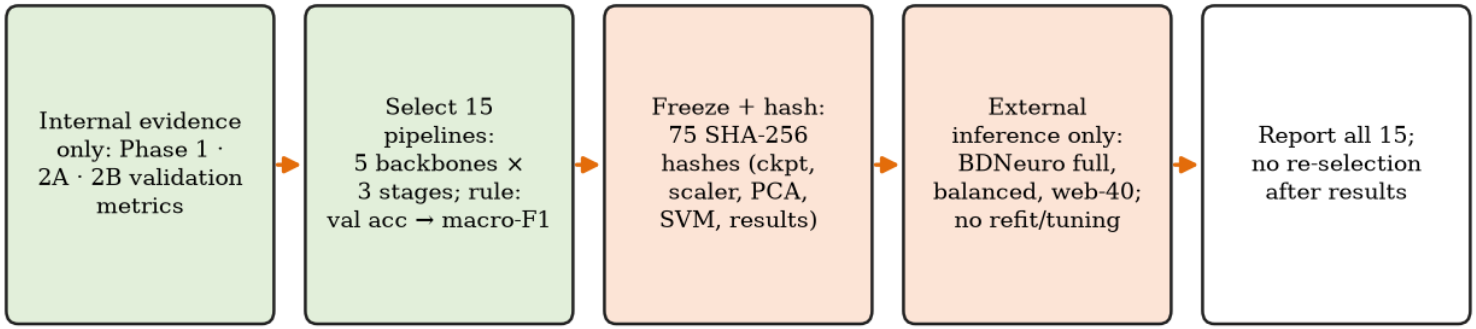
Factor	Mean range
Learning rate	**4.62 pp**
LSTM width	0.58 pp
Dropout	0.42 pp

- Learning rate dominates and is architecture-dependent (frozen VGG ↔ fine-tuned ResNet)
- ResNet50 LR 5e-4: 94.75% at epoch 25 → 90.99% at epoch 30 (−3.76 pp drift)
- Best-neural state vs epoch 30 downstream: 20 better · 8 worse · 7 tied
- Strongest changed configs: VGG16 lr_5e-4 (96.50) · VGG19 dropout_0.5 (95.54) · ResNet50 dropout_0.2 (99.13) · ResNet152 lstm_512 (99.30) · EffNet-B3 lstm_512 (99.39)

Frozen 15-Pipeline Protocol

Selected on internal validation only — frozen and hashed before any external image

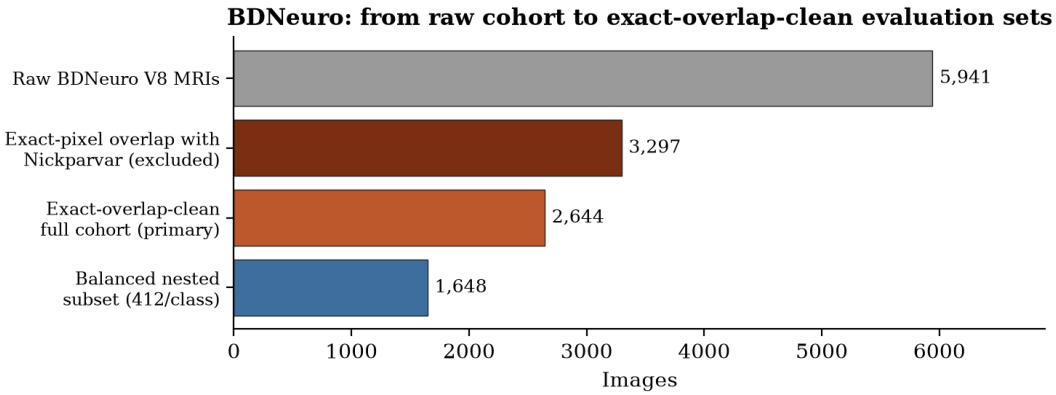
Pre-external freeze protocol (manifest 2026-10-02 16:50 UTC; first external output 2026-10-03 07:54 UTC)



All 75 hashes re-verified against current artifacts during report preparation: 75/75 match.

frozen_15_manifest.json · 15 × 5 = 75 SHA-256 hashes · re-verified 75 / 75 match

BDNeuro: Overlap Audit & Clean Cohort



Audit (vs 7,023)	Count
Exact file pairs	3,421
Exact pixel pairs	3,519
Unique exact-pixel images	3,297
pHash ≤ 6 pairs	9,981
pHash ≤ 4 pairs	6,664

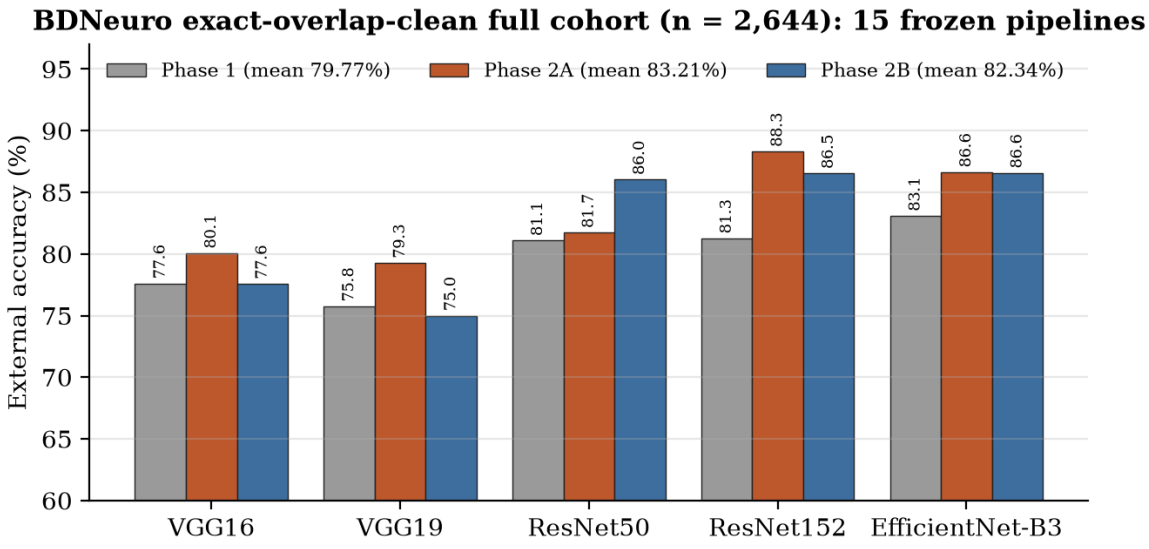
Cohort	Glioma	Mening.	Notumor	Pituit.	Total
Exact-clean (primary)	1,159	472	412	601	2,644
Balanced (nested)	412	412	412	412	1,648

Raw BDNeuro 5,941 is NOT an independent external set.

Rule
Exclude exact decoded-pixel matches only
Keep non-identical pHash candidates (screening, not proof)

BDNeuro External Results (n = 2,644)

All 15 frozen pipelines — no retraining, refitting or recalibration



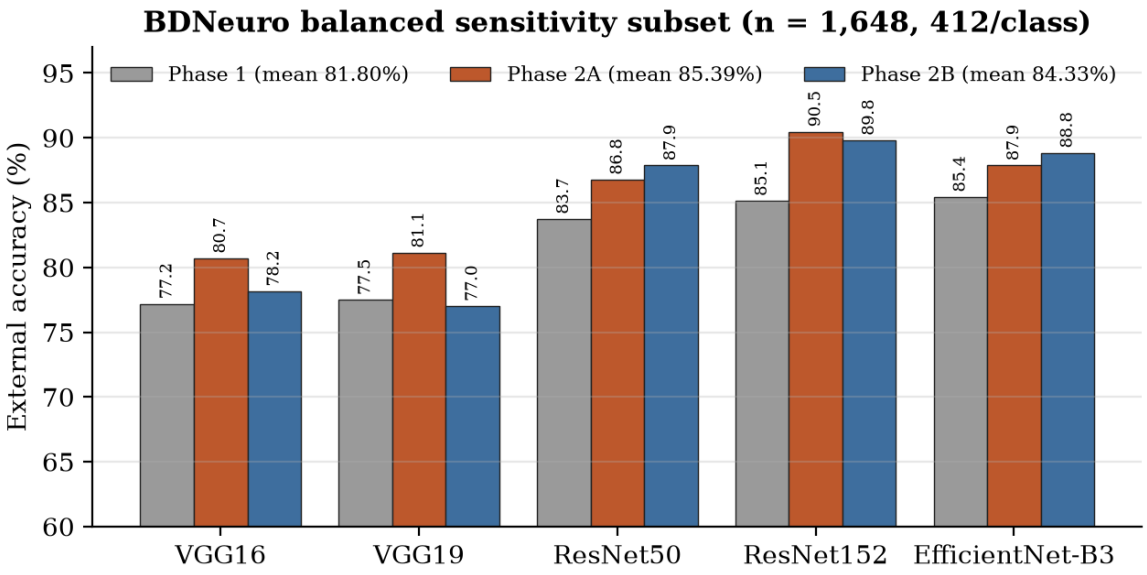
Stage	Mean acc
Phase 1	79.77%
Phase 2A	83.21%
Phase 2B	82.34%

Best: ResNet152 Phase 2A
88.31% · F1 87.85%
gap -10.29 pp

Phase 2A > Phase 2B > Phase 1 on average; Phase 2B gains did not transfer consistently (helped ResNet50, hurt VGG16/19).

Balanced Sensitivity Subset (n = 1,648)

412 per class · nested inside the 2,644 cohort — not a second dataset



Stage	Mean acc
Phase 1	81.80%
Phase 2A	85.39%
Phase 2B	84.33%

Best: ResNet152 Phase 2A
90.47% · F1 90.50%
gap -8.13 pp

Balancing raised accuracy in 14 of 15 pipelines (glioma is 43.8% of the natural cohort and below-average recall for most models).

Web-40 Stress Test — Methodology

Source-verified exploratory stress test — not clinical validation

- 40 images: 10 glioma · 10 meningioma · 10 notumor · 10 pituitary
- Diagnosis from the original source page (search engines only for discovery)
- Reject CT, histology, segmentation maps, ambiguous labels
- Audit vs Nickparvar (7,023) + BDNeuro clean (2,644): exact file, exact pixel, pHash
- Locked before inference; 15 frozen pipelines → 600 predictions

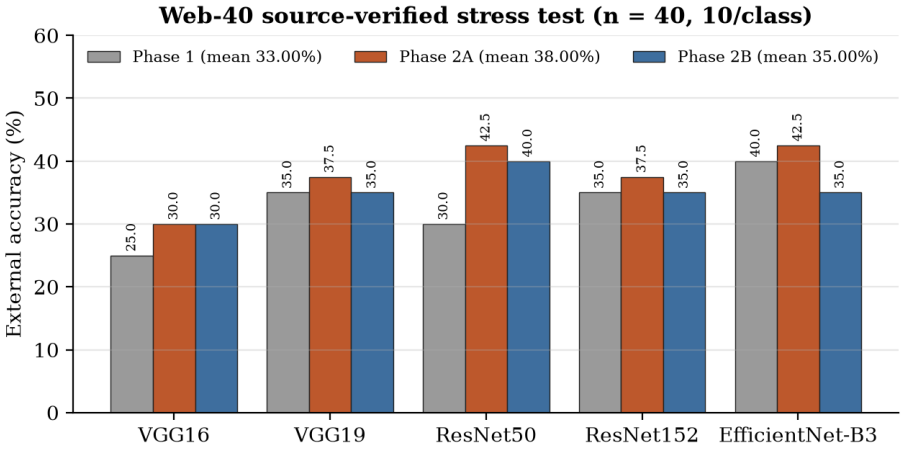
NOR_01 — pHash 0 vs Te-no_0227
same slice visually → rejected & replaced BEFORE inference

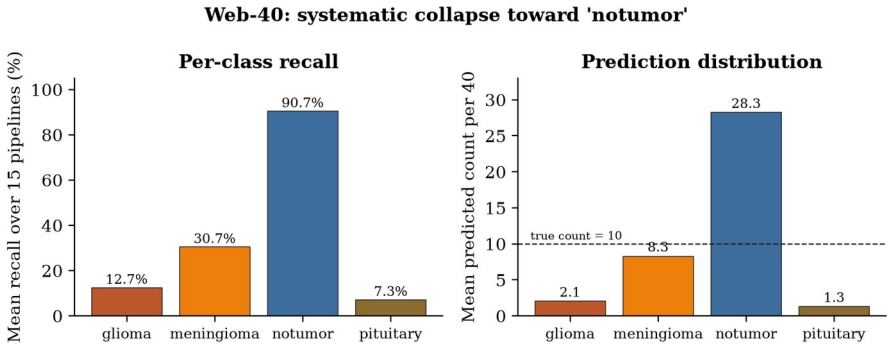
MEN_02 — pHash 6 vs Tr-me_0633
clearly different scan → retained

Final audit
0 exact overlaps · pHash ≤ 4: 0 · LOCKED_BEFORE_INFERENCE

Web-40 Results

Severe domain shift: 25.0–42.5% accuracy and collapse toward notumor



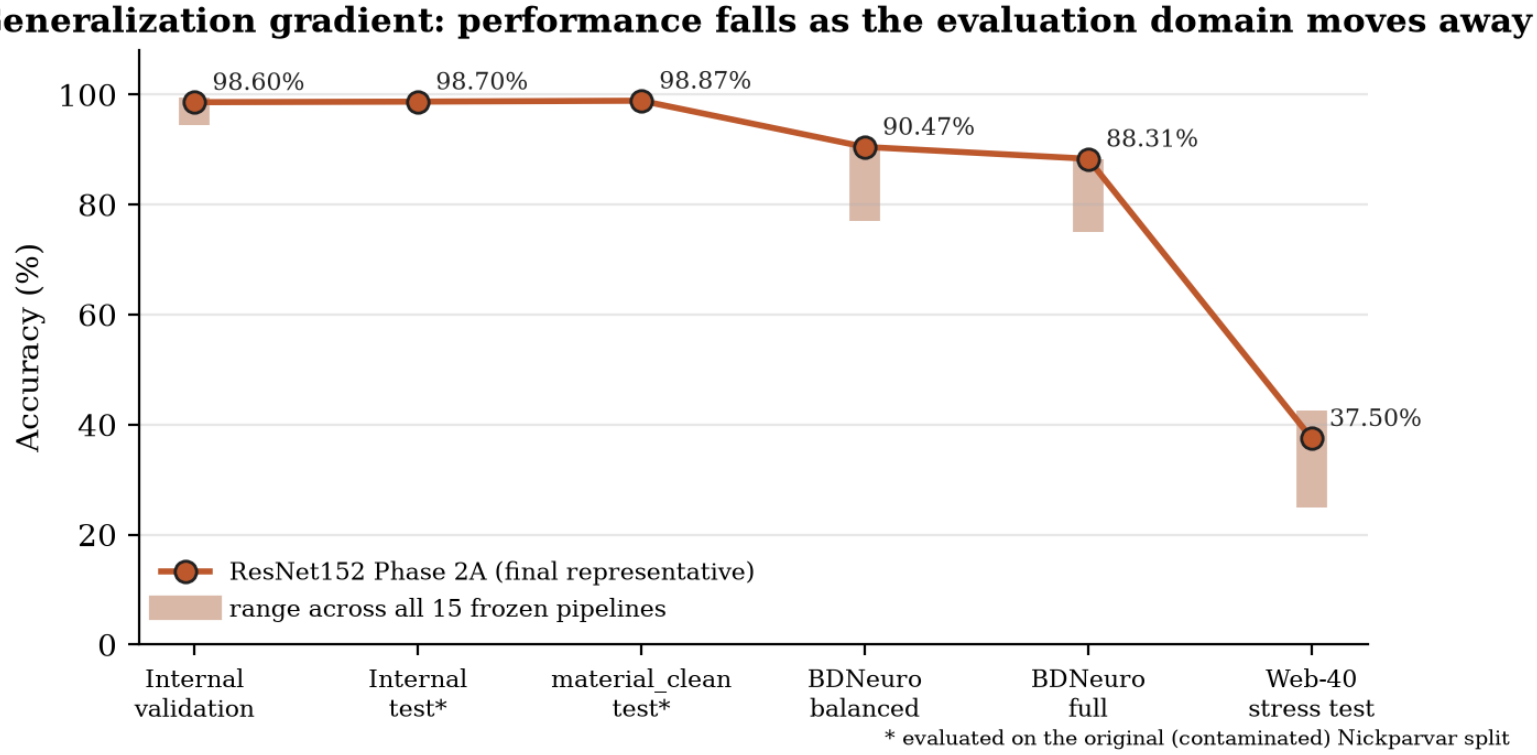


Stage	Mean acc	Mean macro F1
Phase 1	33.00%	26.56%
Phase 2A	38.00%	30.56%
Phase 2B	35.00%	26.95%

Best 42.5% (17/40): ResNet50 2A, EffNet-B3 2A
EffNet-B3 Phase 1 = 40.0% · 1 image = 2.5 pp

Cross-Domain Performance Progression

Internal → material-clean → BDNeuro → web-40



High public-dataset accuracy does not guarantee robust out-of-distribution performance

*

Final Project Representative

ResNet152 · Phase 2A · best_val_within_30 · epoch 10

Evaluation	Accuracy
Internal validation (full hybrid)	98.60%
Internal test*	98.70%
material_clean test*	98.869%
BDNeuro full (2,644)	**88.31%**
BDNeuro balanced (1,648)	**90.47%**
Web-40 stress test	37.5%

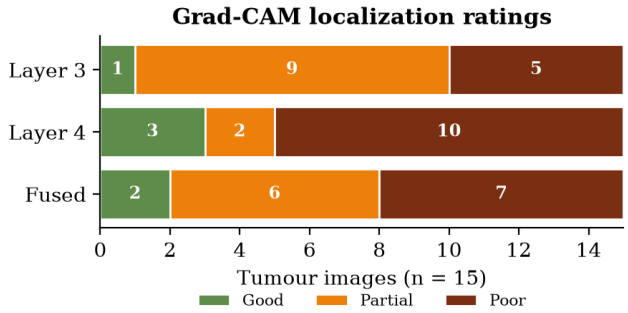
Configuration
RMSprop · LR 1e-4 · LSTM 256 · dropout 0.3 · Dense 128
StandardScaler → PCA 64 → linear SVM (C = 1)

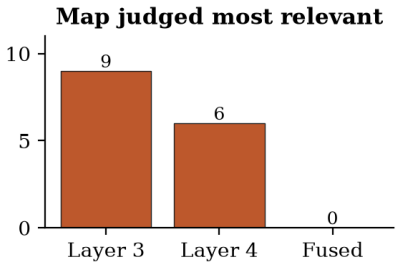
Why
15 frozen on internal validation; representative = strongest structured (BDNeuro) evidence + smallest gaps

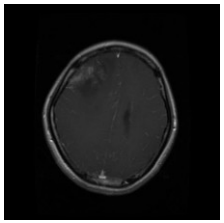
Web-40 was NOT used for selection (37.5%, below two other pipelines). No model was retrained or altered after external results.
* original contaminated split · not claimed to be universally superior

Explainability — Grad-CAM

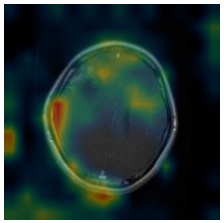
ResNet152 (Phase 1 model) · seed 42 · 20 images, 5/class · Layer 3, Layer 4, fused



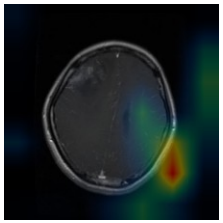




Input (glioma)



Layer 3 · Partial



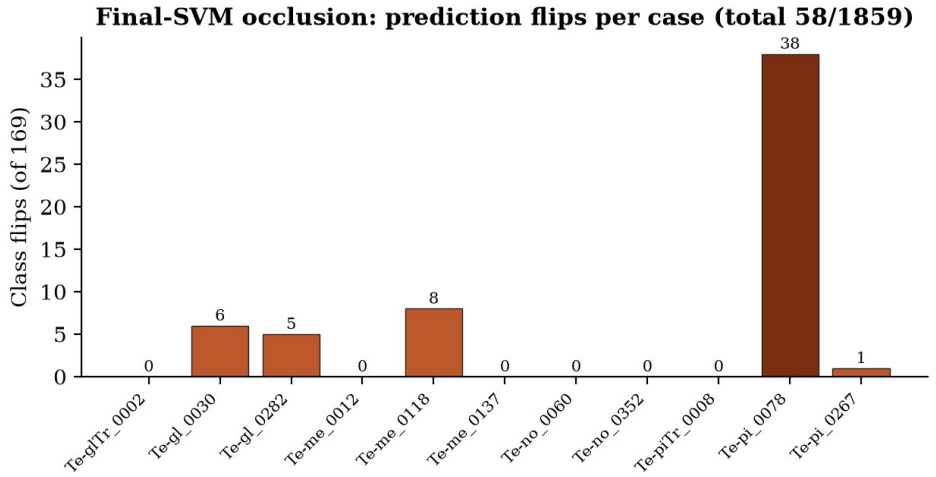
Layer 4 · Poor

Best map: Layer 3 = 9 · Layer 4 = 6 · Fused = 0

- Neural head & SVM agreed 20/20; both correct 19/20 · shared error Te-pi_0078 (pituitary → glioma)
- Correct predictions often had off-lesion maps (skull margin, borders, extracranial)
- Coarse contribution maps — NOT tumour segmentation · Phase 1 ResNet152 only

Explainability — Final-SVM Occlusion

11 cases · patch 32×32 · stride 16 · 169 occlusions/image · 1,859 total

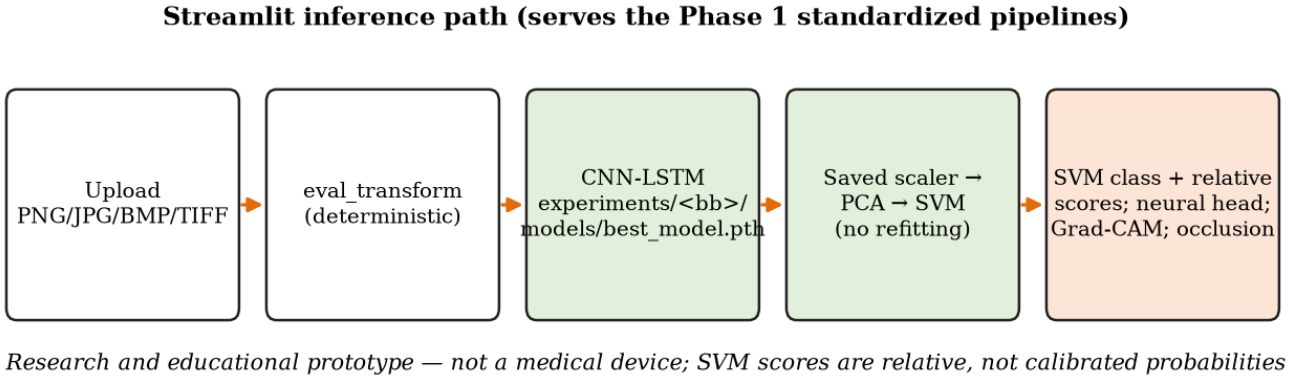


Measure	Value
Tumour rating G/P/Poor	2 / 2 / 5
Grad-CAM better	4
Occlusion better	3
Both off-lesion	2
Flips (pooled)	58 / 1,859 (3.12%)

- Probes the deployed SVM decision directly — but no more lesion-focused than Grad-CAM
- Correct cases stable (median 0 flips); misclassified Te-pi_0078 fragile (38/169 = 22.5%)
- Per-image normalization: read maps with effect sizes · not segmentation

Streamlit Research Prototype

Upload → choose backbone → SVM prediction, neural head, scores, metrics, Grad-CAM, occlusion



- Tabs: Predict MRI · Model performance · Compare models · Methodology
- Saved scaler/PCA/SVM — no refitting on uploads
- Relative scores, not calibrated probabilities · no DICOM

Verified limitation
GUI loads experiments/<backbone>/ → serves Phase 1 pipelines, NOT the final ResNet152 Phase 2A epoch-10 model

Major Findings

- **Hybrid works internally:** 93.5–98.5% (Phase 1 test), up to 99.30% validation (Phase 2A)
- **PCA + SVM ↑ test for all 5 backbones; validation mixed** → downstream-classifier comparison, no universal claim
- **Learning rate dominates sensitivity** (4.62 pp vs 0.58 / 0.42); architecture-dependent
- **Public split has material duplicate leakage** (562 pairs, 250 test images); material_clean stable but post-hoc
- **Internal tuning gains did not reliably transfer:** Phase 2A > 2B externally on average
- **Generalization gradient:** ~99% internal → 75–88% BDNeuro → 25–42.5% web, with notumor collapse
- **XAI:** correct predictions often with off-lesion maps — accuracy ≠ lesion-centred reasoning

Limitations

- Original split contaminated; internal numbers are historical — material_clean does NOT repair training contamination
- Patient-level independence cannot be established (no patient IDs — Nickparvar, BDNeuro or web)
- Single seed / single split; no CIs — Phase 2B differences are not statistically established
- VGG frozen vs others fine-tuned; Phase 1 budgets unequal; no no-LSTM / no-PCA ablation
- BDNeuro balanced subset is nested in the 2,644 cohort; BDNeuro README licence/citation unfilled
- Web-40: n = 40, 1 image = 2.5 pp; exploratory, not clinical
- XAI: small, single-rater, Phase 1 ResNet152 only · GUI serves Phase 1 pipelines
- Research & educational prototype — no clinical or safety claim

Conclusion & Future Work

Conclusion
All roadmap phases (1, 2A, 2B, 2C, 2D, 3, 4, 5, 6) complete, plus a web stress test beyond plan.
Five hybrid pipelines reach ~98% on the public split — but that split leaks, structured external accuracy is 75–88%, and web accuracy falls to 25–42.5%.
High public-dataset accuracy does not guarantee robust out-of-distribution performance.

- Duplicate-group-aware retraining · patient-identified, multi-site data · repeated seeds + CIs
- Ablations (no-LSTM / no-PCA) · MRI-specific normalization & domain adaptation, tested on NEW external data
- Calibration · quantitative XAI with expert masks · GUI → final Phase 2A model · DICOM support

References

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[12] Musthafa et al. (2024), BMC Med. Imaging 24, 107 · [13] Panigrahi et al. (2025), Discov. Appl. Sci. · [14] Sahoo et al. (2024), IJCIS 17 · [15] Kumar et al. (2021), MTAP 80(9)
Datasets: Nickparvar, Brain Tumor MRI Dataset (Kaggle) · BDNeuro-MRI V8 (README; licence/citation to be completed)

Thank you

Questions?

Backup — BDNeuro Full (n = 2,644), all 15

Backbone	Stage	Variant	Int. val %	Ext. acc %	Macro F1 %	Gap pp
VGG16	Phase 1	early_stopping	94.93	77.61	75.50	-17.32
VGG16	Phase 2A	lr1e-4_lstm256_dropout0.3	95.89	80.07	78.71	-15.82
VGG16	Phase 2B	lr_5e-4	96.50	77.57	76.01	-18.93
VGG19	Phase 1	early_stopping	94.49	75.76	74.80	-18.73
VGG19	Phase 2A	lr1e-4_lstm256_dropout0.3	95.63	79.27	78.12	-16.35
VGG19	Phase 2B	dropout_0.5	95.54	75.00	74.27	-20.54
ResNet50	Phase 1	early_stopping	97.55	81.09	79.93	-16.46
ResNet50	Phase 2A	lr1e-4_lstm256_dropout0.3	98.69	81.73	81.42	-16.96
ResNet50	Phase 2B	dropout_0.2	99.13	86.04	84.83	-13.08
ResNet152	Phase 1	early_stopping	98.25	81.28	80.84	-16.97
ResNet152	**Phase 2A**	**lr1e-4_lstm256_dropout0.3**	**98.60**	**88.31**	**87.85**	**-10.29**
ResNet152	Phase 2B	lstm_512	99.30	86.54	85.67	-12.76
EfficientNet-B3	Phase 1	early_stopping	98.51	83.09	81.93	-15.42
EfficientNet-B3	Phase 2A	lr1e-4_lstm256_dropout0.3	99.30	86.65	85.12	-12.65
EfficientNet-B3	Phase 2B	lstm_512	99.39	86.57	85.53	-12.81

Backup — BDNeuro Balanced (n = 1,648), all 15

Backbone	Stage	Variant	Int. val %	Ext. acc %	Macro F1 %	Gap pp
VGG16	Phase 1	early_stopping	94.93	77.18	76.82	-17.74
VGG16	Phase 2A	lr1e-4_lstm256_dropout0.3	95.89	80.70	80.62	-15.18
VGG16	Phase 2B	lr_5e-4	96.50	78.16	77.88	-18.35
VGG19	Phase 1	early_stopping	94.49	77.49	77.64	-17.00
VGG19	Phase 2A	lr1e-4_lstm256_dropout0.3	95.63	81.13	81.24	-14.50
VGG19	Phase 2B	dropout_0.5	95.54	77.00	77.15	-18.54
ResNet50	Phase 1	early_stopping	97.55	83.74	83.80	-13.81
ResNet50	Phase 2A	lr1e-4_lstm256_dropout0.3	98.69	86.77	86.90	-11.92
ResNet50	Phase 2B	dropout_0.2	99.13	87.86	87.87	-11.26
ResNet152	Phase 1	early_stopping	98.25	85.13	85.13	-13.12
ResNet152	**Phase 2A**	**lr1e-4_lstm256_dropout0.3**	**98.60**	**90.47**	**90.50**	**-8.13**
ResNet152	Phase 2B	lstm_512	99.30	89.81	89.76	-9.49
EfficientNet-B3	Phase 1	early_stopping	98.51	85.44	85.60	-13.08
EfficientNet-B3	Phase 2A	lr1e-4_lstm256_dropout0.3	99.30	87.86	87.79	-11.44
EfficientNet-B3	Phase 2B	lstm_512	99.39	88.83	88.96	-10.55

Backup — Web-40 (n = 40), all 15

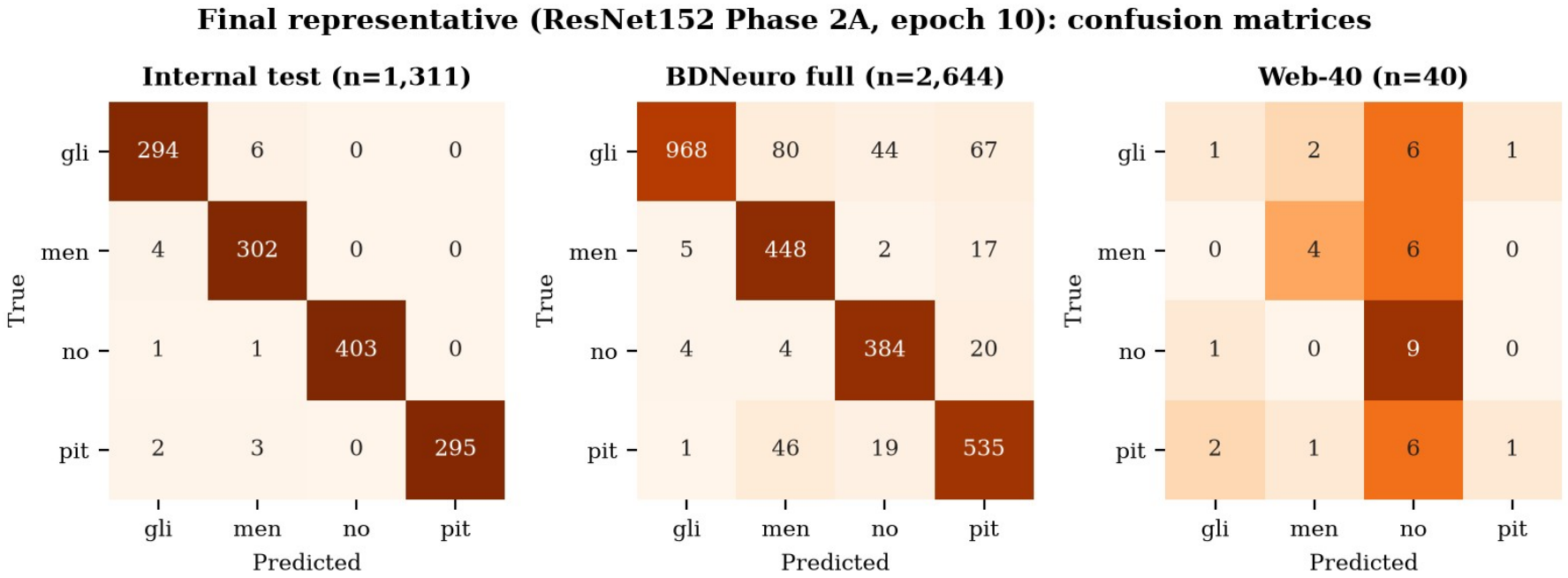
Backbone	Stage	Variant	Int. val %	Ext. acc %	Macro F1 %	Gap pp
VGG16	Phase 1	early_stopping	94.93	25.00	14.58	-69.93
VGG16	Phase 2A	lr1e-4_lstm256_dropout0.3	95.89	30.00	23.23	-65.89
VGG16	Phase 2B	lr_5e-4	96.50	30.00	21.59	-66.50
VGG19	Phase 1	early_stopping	94.49	35.00	28.89	-59.49
VGG19	Phase 2A	lr1e-4_lstm256_dropout0.3	95.63	37.50	27.78	-58.13
VGG19	Phase 2B	dropout_0.5	95.54	35.00	25.45	-60.54
ResNet50	Phase 1	early_stopping	97.55	30.00	26.49	-67.55
ResNet50	Phase 2A	lr1e-4_lstm256_dropout0.3	98.69	42.50	38.18	-56.19
ResNet50	Phase 2B	dropout_0.2	99.13	40.00	33.91	-59.13
ResNet152	Phase 1	early_stopping	98.25	35.00	28.08	-63.25
ResNet152	**Phase 2A**	**lr1e-4_lstm256_dropout0.3**	**98.60**	**37.50**	**31.66**	**-61.10**
ResNet152	Phase 2B	lstm_512	99.30	35.00	25.74	-64.30
EfficientNet-B3	Phase 1	early_stopping	98.51	40.00	34.78	-58.51
EfficientNet-B3	Phase 2A	lr1e-4_lstm256_dropout0.3	99.30	42.50	31.97	-56.80
EfficientNet-B3	Phase 2B	lstm_512	99.39	35.00	28.07	-64.39

Backup — Phase 2B Configurations & Selection

Backbone	Config	State	Epoch	SVM val	SVM val F1
VGG16	lr_5e-4	best_val_within_30	27	96.500%	96.321%
VGG19	dropout_0.5	epoch_30	30	95.538%	95.382%
ResNet50	dropout_0.2	best_val_within_30	28	99.125%	99.122%
ResNet152	lstm_512	best_val_within_30	26	99.300%	99.265%
EfficientNet-B3	lstm_512	best_val_within_30	29	99.388%	99.377%

Backbone	LR range	LSTM range	Dropout range	LSTM512 − base
VGG16	5.77	0.44	0.17	+0.44
VGG19	5.60	1.22	0.52	+0.70
ResNet50	3.85	0.61	0.79	+0.61
ResNet152	4.99	0.44	0.26	+0.44
EfficientNet-B3	2.89	0.17	0.35	+0.09

Backup — Final Model Confusion Matrices



ResNet152 Phase 2A (best_val_within_30, epoch 10): internal test · BDNeuro full · web-40. Web column 'no' shows the notumor collapse.

Backup — Evidence & Artifact References

Result	Repository source
Phase 1	experiments/<bb>/results/{final,neural_head,training}_results.json
Leakage audit	final_project_docs/duplicate_leakage_audit/duplicate_leakage_audit.json
Leakage policies	final_project_docs/leakage_corrected_evaluation/leakage_corrected_summary.csv
Phase 2A	phase2/experiments_30epoch/<bb>/…; phase2/results/leakage_corrected_30epoch/
Phase 2B	phase2/hyperparameter_dl/<bb>/<run>/…
Freeze	phase2/external_validation/frozen_15_manifest.json (75 hashes, 75/75 verified)
BDNeuro	phase2/external_validation/bdneuro_*/ ; comparison_15_models_bdneuro_{full,balanced}/
Web-40	web_stress_40_audit/, web_stress_40_locked/, comparison_15_models_web_stress_40/
XAI	gradcam_evaluation/*.csv ; occlusion_evaluation/*.csv
GUI / augmentation	gui/model_registry.py L51–60 ; src/data_pipeline_torch.py L65–78